

INDUSTRIAL SURFACE DEFECT INSPECTION SYSTEM

Phase 1 — Requirements and Architecture Assessment Report

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1. Official Problem Statement

To build an inspection system that identifies cracks or fractures in a finished material. Manual final inspection is repetitive and may produce poor quality control, additional rejection and factory loss. The challenge is to automate inspection using cameras and machine-learning algorithms.

2. Interpretation and Scope

Term	Interpretation and scope adopted
Finished material	A manufactured product or component at the end-of-line or final-quality-check stage, not a road or civil structure. Target materials: ceramic, steel, painted metal, glass and plastic.
Surface defect	A visible external discontinuity or mark. The operational taxonomy is crack and scratch. Other surface patterns are retained as hard negatives so that the system does not incorrectly reject acceptable products.
Crack	A thin, elongated separation or crack network requiring boundary-level localisation. Segmented as its own class.
Scratch	A thin surface mark that can resemble a crack and is therefore segmented as a separate class to prevent confusion.
Fracture	A severity state, not a separate segmentation class. A crack region whose measured geometry (length, maximum width, or connectivity into a crack network) exceeds the material-specific fracture threshold is reported as a fracture. This keeps the visual model at three classes while still answering the problem statement, which asks for cracks or fractures.
Inspection output	Automated inspection evidence and decision support: pixel mask, defect type, region geometry, overlay image, material-specific accept / reject / review recommendation, and a traceable record.
Automation	Automated visual search producing an accept, reject or review-required result. Human inspectors remain responsible for monitoring, handling uncertain cases, approving thresholds and overriding decisions when justified.

3. Industrial Context and Current Process

Final visual inspection is typically performed after manufacturing, finishing or assembly and before packaging or delivery. Surface appearance varies by material, process, lighting, texture and geometry. A visual system can only identify defects that are optically visible in the captured view; it does not replace ultrasonic, radiographic, penetrant, magnetic-particle or other non-destructive testing when hidden or subsurface flaws matter.

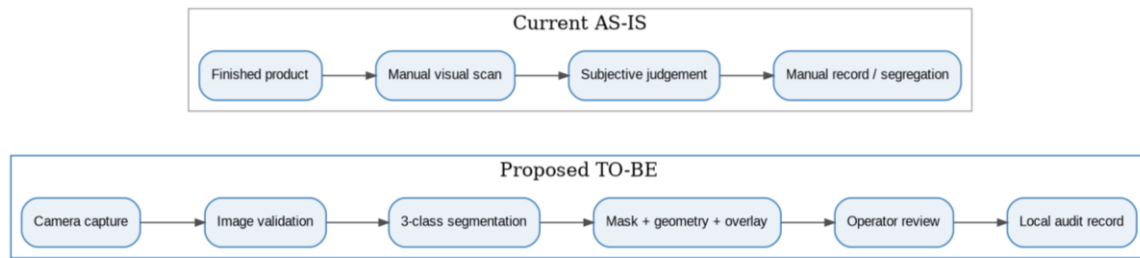


Figure 1. Current manual (AS-IS) workflow and proposed operator-assisted (TO-BE) workflow.

4. Weaknesses of the Current Manual Process

Weakness	Industrial consequence	Requirement response
Fatigue during repetitive visual search	Variable attention and inspection quality across a shift	Automated screening with a short, focused operator review step
Subjective thresholds	Different inspectors may treat the same mark differently	Objective masks and measured geometry; review records preserved
Lighting and reflection	Cracks may disappear, or normal texture may look defective	Image-quality gate, controlled illumination and condition-based tests
High clean-to-defect imbalance	A model can learn the background or over-flag line-like texture	Hard negatives, a false-positive-area metric and frozen clean tests
Weak traceability	Difficult to reproduce a decision or a model state	Image hash, model version, configuration and output logging

5. Stakeholders and Users

Stakeholder	Need	Influence	Key concern
MSME owner / factory manager	Affordable second-check capability with visible benefit	High	Cost, downtime, excessive review or rejection
Quality inspector	Fast evidence that supports a decision	High	False alarms, unclear overlays, awkward workflow

Stakeholder	Need	Influence	Key concern
Quality engineer	Repeatable metrics, traceability, dataset and model governance	High	False negatives, threshold drift, invalid claims
Production operator	Simple capture and clear recapture instructions	Medium	Focus, lighting, camera positioning and line speed
Maintenance engineer	Health checks and reproducible deployment	Medium	Camera, distortion, model or configuration failure
System administrator / ML maintainer	Controlled models, versions and local data	Medium	Unauthorised changes, leakage, unreproducible results
Customer / auditor	Conforming products and defensible records	High	Accountability and evidence quality

The selected primary users are Indian manufacturers handling ceramic, steel & metal based products, glass and plastic products at the end-of-line inspection stage. The system is designed as a low-cost automated inspection and decision-support station that runs on commodity edge compute with local inference. It produces accept, reject or review-required results, while quality personnel monitor operation, approve material-specific thresholds, review uncertain cases and retain authority to override a decision. Public vendor pricing for comparable commercial systems is consultation-based, so no cost-comparison claim is made in this report.

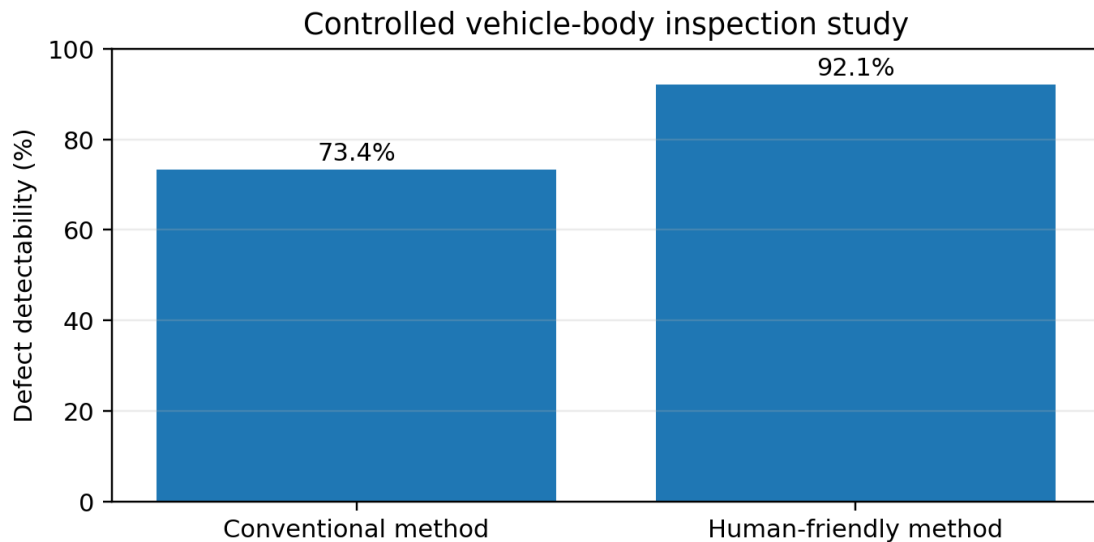
6. Published Surveys and Empirical Evidence

6.1 Fatigue and quality in manufacturing

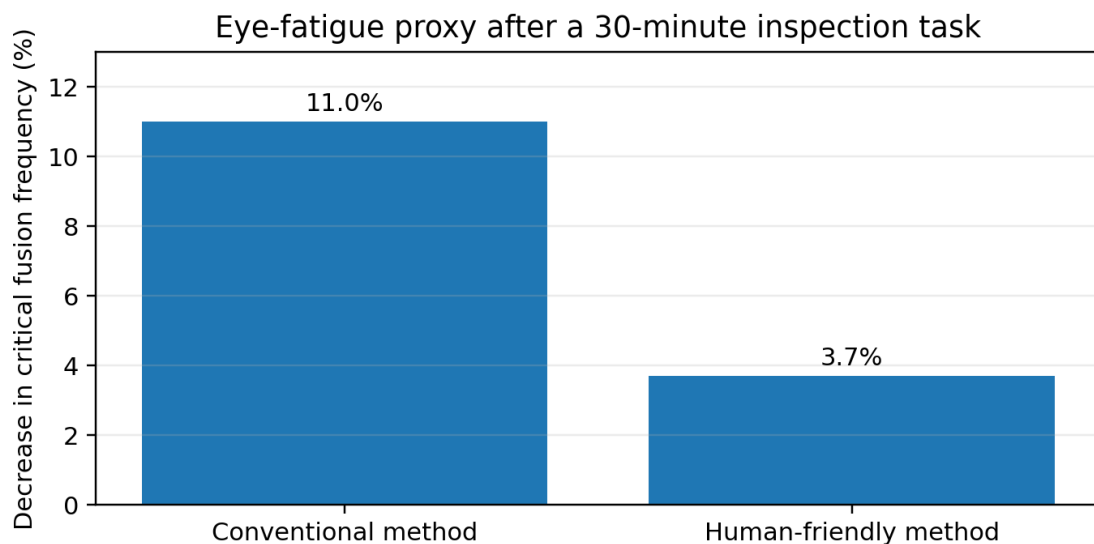
Yung et al. examined the fatigue–quality relationship in manufacturing and reported that fatigue acted as an intermediary between human factors and manufacturing quality, accounting for up to 42% of the variance in quality deficits. This supports treating repetitive inspection workload as a quality-system concern rather than merely an employee-comfort issue [R1].

6.2 Controlled visual-inspection comparison

Tjolleng et al. compared a conventional painted-vehicle-body inspection method with a human-friendly method using projected bright–dark stripes. The new method achieved 92.1% defect detectability against 73.4% for the conventional method. The critical-fusion-frequency decrease after a 30-minute task was 3.7% for the new method and 11.0% for the conventional method, indicating lower eye-fatigue impact. The study is directly relevant to lighting and presentation requirements, although its participant sample was very small and it concerned vehicle-body scars and dents rather than this project's crack and scratch taxonomy [R2].



*Figure 2. Published defect-detectability comparison.
Source: Tjolleng et al., Applied Ergonomics, 2023 [R2].*



*Figure 3. Published eye-fatigue proxy comparison.
A lower decrease indicates less deterioration in the study's fatigue proxy [R2].*

6.3 Quality 4.0 organisational-readiness survey

Antony et al. surveyed 147 senior management professionals from manufacturing and service organisations across America, Asia and Europe. The leading readiness factors were top-management commitment, leadership and organisational culture. Reported non-adoption themes included lack of resources, weak linkage to corporate strategy, limited understanding of benefits, high initial investment, and a belief that existing quality methods already produced adequate results. This supports a low-cost architecture with clear business value, training and change-management requirements [R3].

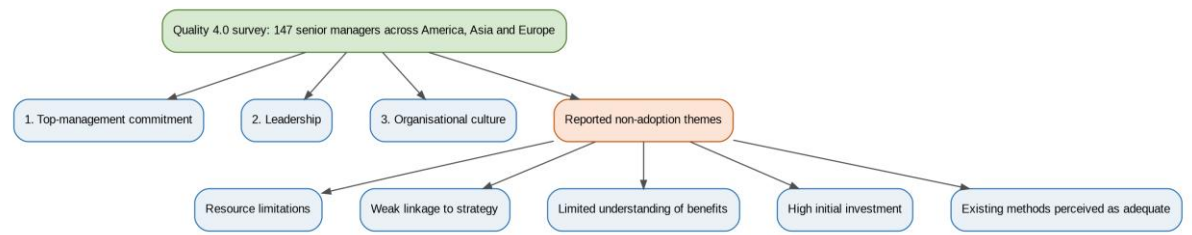


Figure 4. Main findings from the 147-respondent Quality 4.0 readiness survey [R3].

6.4 Indian manufacturing adoption-barrier study

Ghatak and Garza-Reyes analysed fourteen barriers to Quality 4.0 adoption in the Indian manufacturing sector using Interpretive Structural Modelling and MICMAC analysis. The study identified the absence of Quality 4.0 standards and the absence of standardised big-data analytics tools as foundational obstacles. This reinforces the need for transparent requirements, standard metrics, reproducible records, and an architecture that does not assume enterprise data infrastructure [R4].

6.5 Automated visual-inspection research survey

Hütten et al. reviewed 196 open-access publications on deep-learning-based automated visual inspection. Manufacturing use cases represented 31.7% and maintenance use cases 68.3%. The review confirms broad research activity but also shows that a majority of published results come from maintenance rather than factory end-of-line inspection, so direct domain evidence must be separated from adjacent applications [R5].

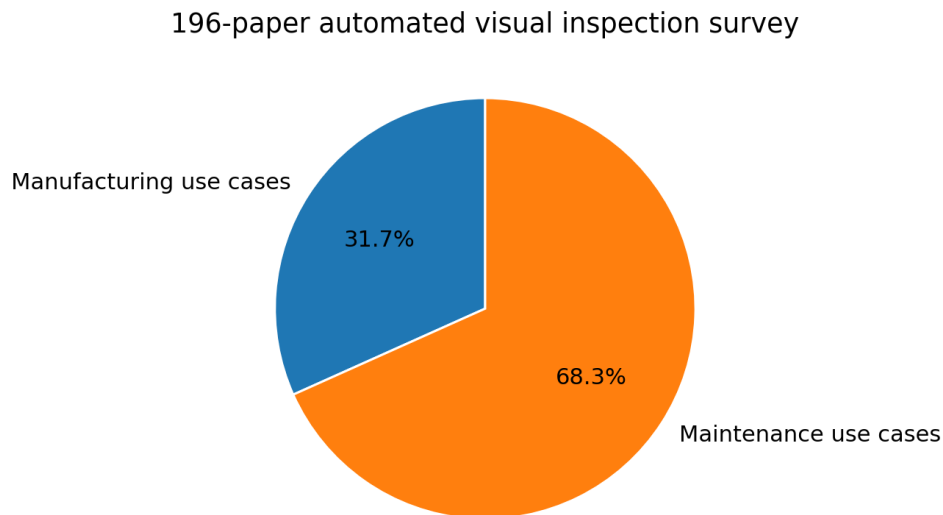


Figure 5. Application distribution in a 196-paper automated visual-inspection survey.

Source: Hütten et al., *Applied System Innovation*, 2024 [R5].

6.6 Evidence-to-requirement implications

Evidence	Supported requirement
Fatigue influences manufacturing quality	Automate sustained visual search; retain human monitoring, exception handling and final accountability

Evidence	Supported requirement
Lighting and presentation materially improve detectability	Treat image acquisition, illumination and material profile as architecture components, not preprocessing details
Leadership, resources and benefit clarity affect adoption	Keep deployment lightweight, offline, explainable and easy to demonstrate
Standards and analytics-tool gaps affect Indian adoption	Use explicit metrics, audit records, versioning and a documented test plan
Research evidence spans different domains	Report factory, transfer-domain and device results separately

7. Functional Requirements

ID	Requirement	Acceptance evidence
FR-01	Accept PNG/JPEG frames from the fixed line-side industrial camera, and stored image directories for batch and regression runs.	Valid input is processed without a network call.
FR-02	Validate format, focus, exposure and field of view before inference.	Invalid or poor capture returns a recapture or invalid state, not a defect result.
FR-03	Normalise valid input to the fixed 256×256 model input while preserving source identity.	Source ID, station ID and processing configuration are retained.
FR-04	Produce a pixel-level mask for every predicted surface-defect region.	Mask dimensions align with the presented source image.
FR-05	Assign every foreground pixel one of three classes: background, crack or scratch.	The emitted label map contains no value outside this taxonomy.
FR-06	Return an empty defect result on a clean surface rather than forcing a prediction.	No foreground region is emitted when all scores remain below validated thresholds.
FR-07	Report area, centreline length and maximum width per region, in pixels and in physical units where camera calibration is available.	Geometry is returned per region and matches reference targets within tolerance.
FR-08	Write an overlay image showing every region and its class.	Overlay can be visually registered to the source image.
FR-09	Process a directory of captures and produce a batch CSV report.	One row per region, plus one row per defect-free image, with failure status where applicable.
FR-10	Log image SHA-256, model version, processing configuration and output for every inspection.	The result can be reproduced exactly from the fixed model and configuration.
FR-11	Require human verification before assisted or pseudo-labelled samples enter training.	The manifest records the reviewer and the acceptance, edit or rejection state.
FR-12	Keep all synthetic and pseudo-labelled samples out of validation and test splits.	The dataset QA gate fails if any non-real sample appears in an evaluation split.

ID	Requirement	Acceptance evidence
FR-13	Display the validation status and known limitations of each material profile.	The interface labels each profile as validated, limited or under evaluation.
FR-14	Allow the operator to select the active material profile from the supported set.	The selected profile and threshold version are stored with the result.
FR-15	Generate an automated accept, reject or review-required decision from defect class, geometry, confidence and material-specific thresholds, including the derived fracture state.	Every valid inspection produces exactly one of the three decision states.
FR-16	Permit an authorised quality inspector to confirm or override an automated decision with a mandatory reason.	User, timestamp, original decision, final decision and reason are logged.
FR-17	Provide an inspection dashboard showing the live result, overlay, material profile, measurements, decision and system status.	All required fields are visible without opening raw logs.
FR-18	Store inspection history and permit filtering by date, material, result, defect type, station and product identifier.	An authorised user can retrieve the matching record set.
FR-19	Generate summary statistics and downloadable inspection reports for a selected batch or time period.	Report totals reconcile with the underlying inspection records.

8. Non-Functional Requirements

ID / attribute	Target	Rationale
NFR-01 Model size	≤ 6 MB float32; ≤ 2 MB after int8 quantisation	Keeps the model deployable on low-cost edge hardware. The selected configuration measures 5.8 MB in float32 ONNX.
NFR-02 Latency	≤ 60 ms median and ≤ 90 ms at the 95th percentile, single-thread CPU at 256×256	The line-side edge unit performs all inference. The selected configuration measures 26.3 ms, leaving headroom for post-processing.
NFR-03 False-positive area	$\leq 0.5\%$ of pixels on clean evaluation images	Directly controls over-flagging and unnecessary operator review.
NFR-04 Material handling	One shared backbone and weight set, with material-specific threshold profiles rather than separate models	Avoids maintaining unrelated models while still allowing validated material-specific decisions.
NFR-05 Offline operation	No network call at inference time	Protects factory images and supports unreliable connectivity.
NFR-06 Determinism	A fixed model and configuration produce identical output for identical input	Enables audit and regression testing.

ID / attribute	Target	Rationale
NFR-07 Evaluation integrity	Frozen, real-image, leakage-tested validation and test splits	Prevents duplicate and synthetic evidence from overstating results.
NFR-08 Thin-structure quality	Report cIDice, tolerant F1 and IoU per class	Pixel accuracy alone is unsuitable for sparse, thin defects.
NFR-09 Explainability	Overlay and region geometry visible to the operator	Supports human judgement and error investigation.
NFR-10 Graceful degradation	Invalid, unknown or weak-domain inputs produce a recapture or review-required state rather than silent acceptance	Limitations must be explicit at the point of use.
NFR-11 Decision safety	No automatic acceptance when image quality, camera health, model configuration or material profile is invalid	A fail-safe review-required state prevents unsupported approval.
NFR-12 Material coverage	Performance reported separately for each material brought into scope	Aggregate accuracy must not hide weak performance on one material.
NFR-13 Availability	Camera, model and storage health exposed before and during inspection	Operators must be able to detect degraded operation.
NFR-14 Usability	A trained operator completes capture review and exception handling in a small, fixed number of interactions	The interface is intended for monitoring rather than continuous manual defect search.
NFR-15 Auditability	Decision, override, threshold and model changes are traceable	Supports quality audits and regression investigation.

9. Deliverables and Expectations

Deliverable	Stated expectation	Project interpretation
Target materials	A clearly bounded inspection domain	Ceramic, steel, metal, glass and plastic finished products, staged by data availability
Defect identification	Detect cracks or fractures using cameras and machine learning	Locate and classify cracks and scratches at pixel level; derive the fracture state from crack-region geometry
Decision output	Support automated quality inspection	Produce accept, reject or review-required using material-specific rules
Human role	Reduce repetitive manual inspection	The inspector monitors operation, handles exceptions, validates thresholds and may override with a recorded reason
Inspection evidence	Clear and reviewable results	Overlay, mask, defect class, geometry, confidence, material profile and audit record

Deliverable	Stated expectation	Project interpretation
Deployment	Practical industrial prototype	Offline CPU inference on a line-side edge unit, with laptop or phone used only for viewing results
Evaluation	Demonstrate accuracy and usability	Material-wise defect metrics, false-positive control, latency, interface workflow and a documented acceptance test plan

10. Material-Specific Requirement Scope

Material	Typical visual challenges	Inspection requirement
Ceramic	Glaze reflection, natural texture, chips, crazing and surface colour variation	Differentiate cracks from glaze patterns and harmless texture; use diffuse illumination and hard negatives
Steel	Scratches, scale, seams, pitting, brushed texture and low-contrast cracks	Prioritise crack-versus-scratch discrimination, thin-structure recall and low false-positive area
Metal	Specular paint, colour variation, dust, dents, paint scratches and reflections	Control lighting, preserve colour information, and separate coating damage from structural crack evidence
Glass	Transparency, refraction, reflections, edges, dust and background leakage	Use a controlled background and illumination, multi-angle capture where necessary, and a fail-safe review state for uncertain cracks
Plastic	Mould lines, glossy or matte finishes, sink marks, colour variation and stress whitening	Differentiate mould seams and cosmetic marks from cracks; collect product-specific real images

11. Enhancements and Their Merit

ID	Enhancement	Description	Merit
E-01	Material inspection profiles	Material-specific thresholds and separate evaluation for each material in scope	Prevents a single generic threshold from hiding domain differences
E-02	Three-class pixel segmentation with derived fracture state	Background, crack and scratch masks, with fracture derived from crack-region geometry, instead of image-level classification	Provides exact location and geometry for explainable decisions while keeping the model small
E-03	Automated accept / reject / review decision	Combines defect class, geometry, confidence and material rules	Fulfil the factory automation objective while retaining a safe uncertainty state

ID	Enhancement	Description	Merit
E-04	Human monitoring and override workflow	The inspector supervises results and records justified overrides	Reduces repetitive work without removing accountability
E-05	Image-quality gate	Rejects blurred, overexposed, incomplete or invalid captures before inference	Prevents false acceptance caused by poor imaging
E-06	Defect geometry analysis	Area, centreline length, maximum width and region count per defect	Enables transparent thresholding, the derived fracture state and future severity rules
E-07	Hard-negative intelligence	Seams, scratches, reflections, mould lines, grain, scale and texture explicitly trained as non-cracks	Controls costly over-rejection
E-08	Offline edge inference	No network call at any point during inspection	Protects factory images and supports poor connectivity
E-09	Quantised deployment with float32 reference check	int8 ONNX model deployed at the line, with a float32 reference model used to verify sensitivity has not shifted	Achieves the size and latency targets without silently losing thin-defect recall
E-10	Traceable inspection records	Image hash, model version, thresholds, material profile and decision history	Supports audit, reproducibility and quality investigation
E-11	Batch dashboard and reports	Live result plus batch statistics, filters and CSV or PDF export	Makes the prototype useful beyond a single-image demonstration
E-12	Human-verified assisted annotation	Foundation-model-assisted mask proposals reviewed by a human before entering training	Expands scarce factory data without treating pseudo-labels as ground truth

12. High-Level Architecture and Design Methodology

12.1 Overview

The system performs automated visual inspection of finished products on a factory conveyor line, using a fixed industrial camera mounted above the belt. It segments the surface into crack and scratch regions at pixel level, measures the geometry of each region, and produces one structured record per item.

The design methodology is requirements-first and evidence-gated. Requirements were derived from the problem statement and the published evidence in Section 6, expressed as testable functional and non-functional statements, and each is traced to an acceptance test in Section 13.5. Architectural choices were then made by measured comparison rather than assumption: five encoder–decoder configurations were exported to ONNX and benchmarked on parameter count, model size and single-thread CPU latency before one was selected.

The system automates the search step of inspection — scanning the full surface of every product for possible defects — which is the part of manual inspection that causes the most eye

fatigue when performed continuously. It does not automate the judgement step by default: a flagged region and its geometry are handed to a human for confirmation, because validating an already-located candidate region is a much smaller visual and cognitive load than searching an entire surface unaided. Automatic validation of flagged regions is a capability the architecture can support, but it is treated as a separate, independently validated extension rather than default behaviour.

Inference runs on local edge compute at the line with no network dependency. Results are viewed separately through a laptop or phone interface, which is read-only and is not used for capture.

12.2 End-to-end system pipeline

A fixed industrial camera captures each product as it passes the inspection zone. The frame is decoded and tagged with product, line and station metadata, resized and normalised to the fixed 256×256 model input, and passed to the segmenter. The three-class output is thresholded, split into connected components per class, and measured by the geometry engine. Results are assembled into one structured record per product and written to three destinations: an append-only audit log, a batch CSV report and an overlay image. Batch processing runs this same pipeline once per stored image; it is not a separate implementation.

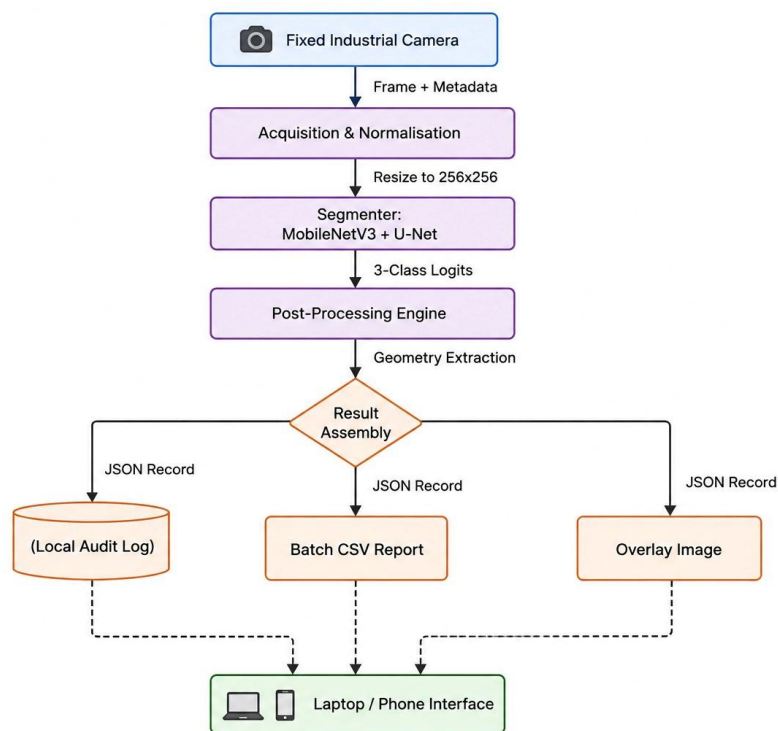


Figure 6. End-to-end inspection pipeline from camera capture to operator-facing outputs.

12.3 Material considerations and data availability

None of the components described in this section — the segmentation network, the post-processing pipeline or the model configuration — depend on a specific target material. Moving to a new material is a matter of retraining on material-specific data and calibrating a threshold

profile, not restructuring the pipeline. The table below records the trade-off that determines the order in which materials are brought into validated scope.

Material	Segmentation-grade public data	Product type	Indian manufacturing base
Steel	Extensive (Severstal, KolektorSDD2, NEU-DET, GC10-DET, DAGM)	Finished sheet and coil product	Large
Ceramic	Limited; mostly classification or lab-condition data	Tile, sanitary ware, tableware	Large
Plastic	Limited; mostly anomaly detection rather than segmentation (MVTec AD, VisA, BTAD, MPDD)	Finished moulded product	Large
Painted metal	Limited; largely proprietary	Finished assembled product	Present
Glass	Very limited; largely proprietary	Finished product	Present
Concrete (reference only, not a target material)	Extensive (CFD, SDNET2018, DeepCrack, Crack500)	Installed structure, not a shipped product	Large

Steel is therefore the first material taken to full validation, because it is the only target material with abundant pixel-annotated public data. The remaining four are staged behind pilot image capture, and each is reported separately under NFR-12 rather than folded into an aggregate score.

12.4 Defect classes

Class	Typical shape	Typical width	Notes for detection
Crack	Thin, often curved or branching linear structure	Narrow — a few pixels	Low contrast depending on surface finish; benefits most from full-resolution skip connections in the decoder
Scratch	Linear surface mark, generally straighter than a crack	Wider than a crack on average	Higher contrast, but can be confused with surface reflections and brushed texture

Both are line defects rather than area defects. Fracture is not a third mask class: it is a severity state derived at the decision layer from a crack region whose measured length, maximum width or network connectivity exceeds the material-specific fracture threshold. This keeps the segmentation head at three classes — background, crack, scratch — while still producing the crack-or-fracture output the problem statement asks for, and it avoids asking the model to learn a boundary between crack and fracture that is in practice a threshold decision rather than a visual one.

Broader area and texture anomalies such as pin holing, crazing and discoloration are out of scope. Including them would require the model to learn a materially different visual signature, diluting the capacity and annotation effort directed at crack and scratch detection.

12.5 Image acquisition

A fixed industrial camera captures each product at a constant working distance under constant lighting as it passes through the inspection zone, producing one frame per product with a timestamp and station identifier. Because the setup is fixed rather than handheld, no per-device capture branching or distortion correction is required downstream, and calibration between pixels and physical units needs to be established only once per station.

12.6 Model architecture

Component	Role
MobileNetV3-Small encoder (ImageNet-pretrained)	Extracts features at multiple resolutions; chosen for low parameter count and CPU latency relative to larger backbones
Lightweight U-Net decoder (channel widths 64, 48, 32, 24, 16)	Reconstructs a full-resolution segmentation mask from encoder features
Full-resolution skip connections	Carry early, high-resolution encoder features into the decoder so structures only a few pixels wide are not lost at deeper strides
Output head	Produces a three-class per-pixel score: background, crack, scratch

Configuration comparison. All five configurations were exported to ONNX and timed on a single CPU thread at 256×256 input.

Configuration	Parameters	Model size (ONNX, float32)	CPU latency (single thread)
Lightweight custom network (no pretrained backbone)	0.14 M	0.6 MB	13.3 ms
MobileNetV3-Small + lightweight U-Net (selected)	1.43 M	5.8 MB	26.3 ms
MobileNetV3-Large + lightweight U-Net	3.74 M	15.1 MB	34.5 ms
MobileNetV3-Small + full U-Net decoder	3.59 M	14.4 MB	55.4 ms
EfficientNet-B0 baseline	6.25 M	23.4 MB	71.0 ms

The custom network without a pretrained backbone underfitted both defect classes in early experiments and was not carried forward despite its smaller size. MobileNetV3-Small with the lightweight decoder gives the best balance of size, latency and accuracy on thin defect structures, and satisfies NFR-01 and NFR-02 with margin.

Training uses binary cross-entropy combined with a soft Dice loss per class. Class weights are set from measured defect geometry: cracks are consistently narrower and occupy a smaller fraction of foreground pixels than scratches, so without weighting the network would minimise error largely by getting the scratch class right while under-representing crack.

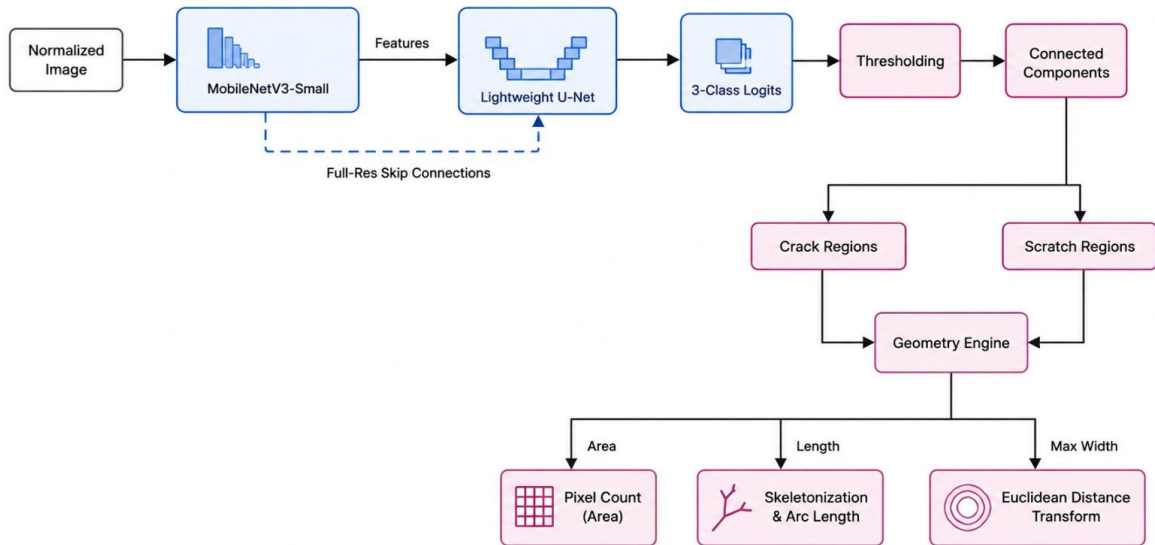


Figure 7. Segmentation network and post-processing geometry engine.

12.7 Post-processing and geometry extraction

The per-pixel output is converted into a region-level result. Per-class thresholding produces an initial label map. Connected components are computed separately for crack and scratch, so adjacent defects of different types stay distinct. Each region's area, length and maximum width are then measured: length by skeletonisation and arc length along the medial axis, and maximum width by the Euclidean distance transform along the skeleton, doubled, so that the widest point is reported rather than an area-over-length average. Regions below a minimum pixel count are discarded to suppress noise from surface glare and texture on otherwise clean products. The resulting geometry is what the decision layer compares against the active material threshold profile, including the crack-length and crack-width thresholds that promote a crack region to the fracture state.

12.8 Deployment and interfaces

Component	Role
Edge inference unit (line-side)	Runs the quantised int8 model through ONNX Runtime, producing one record per captured frame within the line's throughput budget
Float32 reference model	Checks that quantisation has not shifted detection sensitivity, particularly for defects near the minimum detectable size
Append-only audit log	Local, tamper-evident record of every inspection, written before any result is displayed
Batch aggregator	Concatenates per-image records into CSV reports for a run or a time period
Image overlay engine	Renders masks and class labels onto the source image for operator review
Laptop / phone interface	Displays overlays, batch reports and audit entries read from the edge unit's output; read-only, and not used for capture

Every inference run records the model version, an image hash and the line and station identifier, so any result can be traced back to the exact model and configuration that produced it.

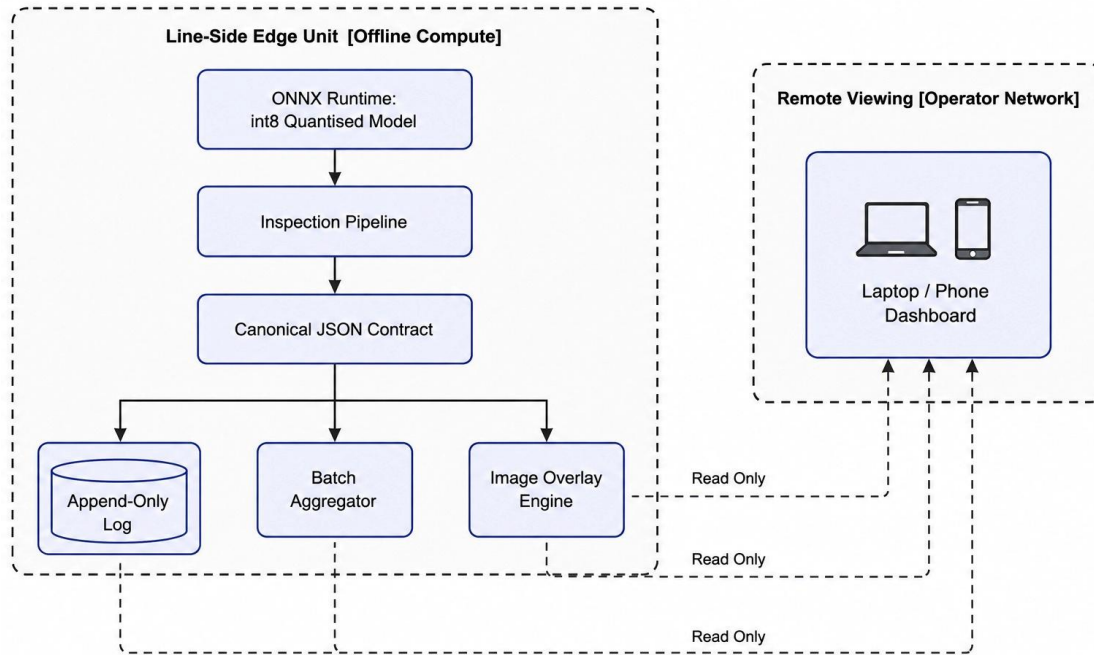


Figure 8. Deployment topology — line-side edge unit with read-only remote viewing.

12.9 Batch processing and audit logging

Given a directory of stored images, the pipeline runs once per file and concatenates results into a single CSV report, with one row per detected region plus one row per defect-free image. Every processed record is also appended to a local, append-only audit log carrying the image hash and model version, giving a traceable inspection history without any network dependency. This same path is used for regression runs against the frozen test split.

12.10 Human judgement and validation

Model scores are used internally for thresholding and ranking candidate regions, but are not shown to the operator as a displayed probability. What reaches the operator is a flagged region with its measured geometry, not a raw score. The reasoning for keeping judgement primarily human is about where the visual and cognitive load actually falls: unaided search across a whole product surface, repeated continuously, is what produces operator eye fatigue, whereas confirming or dismissing a small number of already-located, cropped candidate regions is a substantially lighter task. The system is built to take on the search load and leave the lighter validation load to the operator by default.

Automatic validation — the system deciding that a flagged region is a genuine defect without operator confirmation — is a capability the architecture can be extended to support, since the scores and geometry it needs already exist in the pipeline output. It is treated as a separate, independently validated capability rather than default behaviour, because deploying it before flagged-region accuracy is characterised would shift real judgement onto an uncalibrated score.

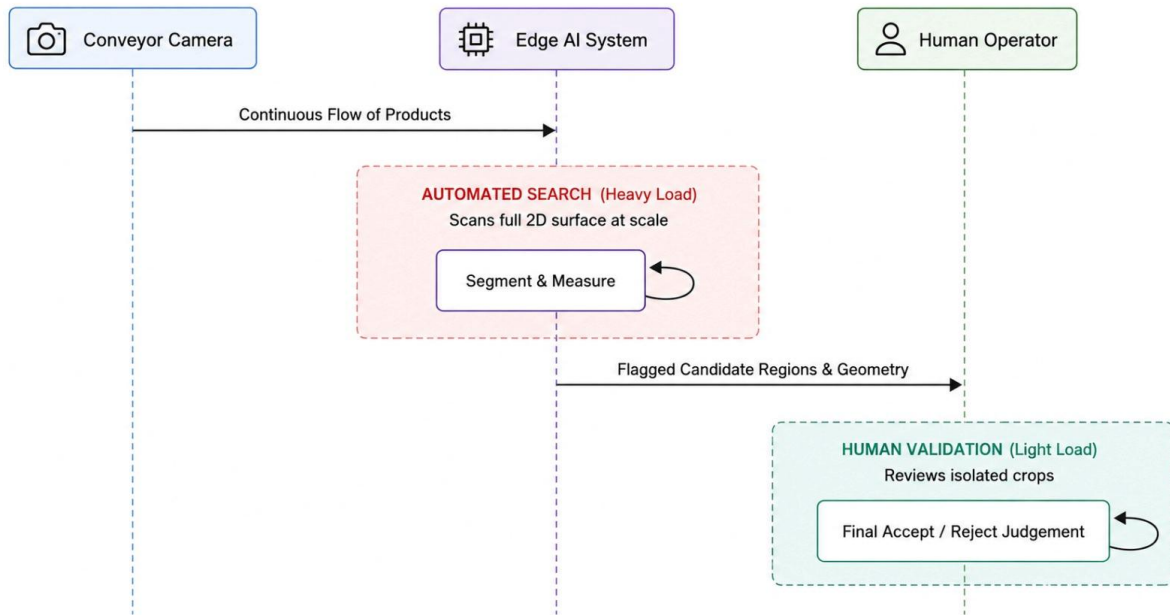


Figure 9. Division of load between automated search and human validation.

12.11 Tool and technology selection

Layer	Selection	Reason
Language	Python 3.11	Mature computer-vision and deep-learning ecosystem; the whole team is fluent in it
Training framework	PyTorch	Straightforward custom encoder–decoder construction and export path to ONNX
Inference runtime	ONNX Runtime (CPU execution provider)	Framework-independent, small footprint, supports int8 quantisation, runs offline
Classical vision and geometry	OpenCV and scikit-image	Connected components, skeletonisation and distance transform are available and well tested
Interface	Lightweight local web application served by the edge unit	Runs in any browser on a laptop or phone with no client installation, and keeps the interface read-only by construction
Storage	Local filesystem with append-only JSON Lines log and CSV export	No database server to administer in an MSME setting; trivially portable and auditable
Version control and tracking	Git, with a dataset manifest and model card per release	Ties every result to a specific code state, dataset split and model version

13. UI and Acceptance Planning

13.1 Interface scope and design principles

The interface serves monitoring and exception handling, not continuous manual defect search. Four principles follow from the requirements. First, the operator sees evidence rather than scores: an overlay and measured geometry, never a raw probability. Second, the default state

is visible at a glance — the decision, the material profile and system health are on screen without navigation. Third, the interface is read-only with respect to inspection results; the only write action available to an operator is a decision override, which requires a reason. Fourth, every screen is legible on a phone as well as a laptop, since the viewing device is not fixed.

13.2 Screen inventory

Screen	Purpose	Key elements	Primary user
Live inspection view	Show the result for the item currently passing the station	Overlay image, decision banner (accept / reject / review), defect class and geometry table, active material profile, camera and model health strip	Production operator
Region review panel	Confirm or dismiss individual flagged regions	Cropped candidate region, measured length and width, accept and reject controls, mandatory reason field on override	Quality inspector
Batch run and report	Run the pipeline over a stored image directory and export results	Directory selector, progress and failure counts, summary statistics, CSV and PDF export	Quality engineer
Inspection history	Retrieve past inspections	Filters for date, material, result, defect class, station and product ID; result list with drill-down to the original overlay	Quality engineer, auditor
Material profile and thresholds	View and approve material-specific thresholds	Threshold values per material, validation status label, version number and change history	Quality engineer
System health and diagnostics	Confirm the station is fit to inspect	Camera connectivity and focus check, model version and hash, storage capacity, last self-test result	Maintenance engineer
Audit log viewer	Inspect the traceable record	Append-only entries with image hash, model version, configuration, decision and override history; export for audit	Auditor, administrator

13.3 Primary interaction flow

The operator's normal path is short by design. A product passes the station and the live inspection view updates with an overlay and a decision banner. If the decision is accept or reject, no action is required and the item proceeds. If the decision is review-required, the region review panel presents the cropped candidate regions one at a time with their measurements; the operator confirms or dismisses each, and on any disagreement with the automated recommendation supplies a reason, which is written to the audit log with the user and timestamp. At the end of a shift or batch, the quality engineer opens the batch report, exports the CSV, and reviews the exception list. The health strip is visible throughout; if the camera, model or storage check fails, the station holds in review-required and no automatic acceptance is issued.

13.4 Datasets

Training and evaluation draw on public segmentation-grade defect datasets, a synthetic generator built by the team, and a pilot capture set from the target line. Public datasets are used under their stated licences; several are restricted to non-commercial research use, which is acceptable for a prototype but is recorded here because it constrains any later commercial deployment.

Dataset	Content	Annotation	Role in this project
Severstal Steel Defect Detection (Kaggle)	12,568 training and 1,801 test images of steel sheet at 1600×256, four defect classes	Run-length-encoded pixel masks	Primary training and internal validation source for the steel profile
KolektorSDD2	3,335 images of production items at approximately 230×630, including scratches and spots	Fine-grained pixel masks	Real production imagery for training and validation; CC BY-NC-SA 4.0, non-commercial
NEU-DET	1,800 grayscale hot-rolled steel strip images at 200×200, six defect classes, 300 per class	Bounding boxes	Hard-negative mining and class-balance reference; not used for mask supervision
GC10-DET	3,570 grayscale steel sheet images, ten defect types from a real production line	Bounding boxes	Hard negatives for surface patterns that must not be flagged as cracks
DAGM 2007	Ten classes of artificially generated textured surfaces with weakly labelled defects	Weak (ellipse) labels	Texture robustness and false-positive control on patterned backgrounds
MVTec AD	5,354 images across 15 object and texture categories, 73 defect types	Pixel-precise ground truth for anomalous regions	Cross-material generalisation checks, including tile and leather textures; non-commercial research licence
Magnetic Tile Defect	1,344 images across six categories including crack and break	Pixel masks	Thin-defect recall on a ceramic-like surface; supports the ceramic profile
DefectForge (team-built synthetic generator)	Parametric crack and scratch synthesis composited onto clean substrate images	Exact masks by construction	Training augmentation only; excluded from validation and test by FR-12
Pilot capture set (team-collected)	Frames from the fixed-camera rig on the target line	Human-annotated masks, reviewed under FR-11	Frozen real test split and final acceptance evidence

Split and governance rules.

Rule	Implementation
Split ratio	70% training, 15% validation, 15% test, stratified by material and by defect presence
Real images only in evaluation	Validation and test contain real images exclusively; synthetic and pseudo-labelled samples are confined to training

Rule	Implementation
Frozen test split	The test split is frozen at the start of Phase 2 and opened only for final acceptance measurement
Leakage audit	A perceptual-hash comparison runs across all three splits before every evaluation; any near-duplicate crossing a split boundary fails the run
Human verification	Assisted or pseudo-labelled masks enter training only after a named reviewer accepts, edits or rejects each one; the manifest records the outcome
Clean-surface set	A dedicated defect-free set is held separately to measure false-positive area under NFR-03
Per-material reporting	Metrics are reported per material and never aggregated into a single headline number

13.5 Acceptance test plan

Each test below is traced to the requirement it discharges. A test passes only if its stated criterion is met on the frozen evaluation data; tests marked critical must all pass before Phase 2 is considered complete.

ID	Traces to	Method	Pass criterion
AT-01	FR-01, FR-03	Feed 200 valid frames from the fixed camera rig	All frames decoded, normalised to 256×256, source and station ID retained
AT-02	FR-02, E-05	Submit 40 deliberately degraded captures (defocus, over- and under-exposure, partial product, occlusion)	At least 95% routed to recapture or invalid; none returns a defect verdict
AT-03	FR-04	Compare mask dimensions with the presented source for every output	Mask registers to the displayed source in 100% of cases
AT-04	FR-05	Run the frozen real test split and inspect the emitted label map	No label outside background, crack, scratch is emitted
AT-05	FR-06, NFR-03	Run the held-out clean-surface set	False-positive area at or below 0.5% of pixels; at least 98% of clean images return an empty defect result
AT-06	FR-07	Measure printed reference targets of known length and width at the calibrated working distance	Length within $\pm 5\%$ and maximum width within ± 1 pixel of ground truth
AT-07	FR-08, NFR-09	Visual inspection of overlays for a 50-image sample	Every flagged region is visible and labelled by class; overlay registers to the source
AT-08	FR-09	Batch run over a 500-image directory	One row per region plus one per clean image; failures carry explicit status; run completes unattended
AT-09	FR-10, NFR-06	Re-run the same 100 images twice with the model and configuration fixed	Byte-identical masks and identical CSV values; hash, model version and configuration present in every record

ID	Traces to	Method	Pass criterion
AT-10	FR-11, FR-12	Inject synthetic and unreviewed pseudo-labelled samples, then run the dataset QA gate	Gate fails and names the offending files; unreviewed samples blocked from the training manifest
AT-11	FR-13, FR-14, NFR-12	Switch the material profile across the supported set and inspect the interface	Selected profile, its validation status and threshold version shown and stored with each result
AT-12	FR-15, NFR-11	Run a mixed set spanning clean, marginal and clearly defective items, including invalid preconditions	Every valid inspection yields exactly one of accept, reject or review-required; invalid preconditions force review-required
AT-13	FR-16, NFR-15	Attempt an override without a reason, then with one	Override blocked until a reason is supplied; user, timestamp, original and final decision logged
AT-14	FR-17, NFR-14	Timed walkthrough with five trained users	All required fields visible without opening raw logs; review of a flagged item completed in no more than three interactions
AT-15	FR-18	Query history by date, material, result, defect class, station and product ID	Returned set matches an independently computed expected set
AT-16	FR-19	Generate a batch report for a selected period	Report totals reconcile exactly with the underlying records
AT-17	NFR-01, NFR-02	Measure exported int8 ONNX size and single-thread CPU latency at 256×256 over 500 runs	Size at or below 2 MB int8 and 6 MB float32; median latency at or below 60 ms and 95th percentile at or below 90 ms
AT-18	NFR-05, E-08	Run a full inspection cycle with all network interfaces disabled	Cycle completes; no outbound connection attempted
AT-19	NFR-07	Run the perceptual-hash leakage audit across all splits	Zero near-duplicate crossings; zero synthetic or pseudo-labelled items in validation or test
AT-20	NFR-08	Evaluate on the frozen real test split	cIDice, tolerant F1 and IoU reported per class; crack-class cIDice meets the Phase 2 target agreed at the M2 gate
AT-21	NFR-10, NFR-13	Disconnect the camera and corrupt the configuration mid-run	Degraded health reported and the station holds in review-required; no silent acceptance
AT-22	E-09	Compare int8 and float32 outputs on defects near the minimum detectable size	Recall drop from quantisation stays within the agreed tolerance, or the float32 model is deployed instead

14. Execution Plan and Schedule

14.1 Approach

Execution is organised into eight one-week work packages, each ending in a named deliverable, with five gate reviews at which work does not proceed until the gate criterion is met. Week numbers are relative to the start of Phase 2 rather than fixed calendar dates, so that the plan can

be aligned to the organisers' milestone dates without restructuring. The team is two people, so the plan deliberately keeps a single active workstream per person and avoids parallel branches that would need integration effort neither member has spare capacity for.

14.2 Work breakdown and schedule

Hour	Work package	Key tasks	Deliverable
H1	WP-1 Dataset consolidation and governance	Assemble public corpora, define and stratify splits, build the perceptual-hash leakage audit, freeze the test split, record licences	Dataset manifest and split-freeze report
H2	WP-2 Data pipeline and annotation	Normalisation and augmentation pipeline, DefectForge synthesis, assisted annotation with human verification loop, hard-negative curation	Reproducible data loader and annotation manifest
H3	WP-3 Baseline three-class model	Train MobileNetV3-Small plus lightweight U-Net, set class weights from measured geometry, evaluate cIDice, tolerant F1 and IoU	Baseline checkpoint and metric report
H4	WP-4 Post-processing and geometry engine	Per-class thresholding, connected components, skeletonisation, Euclidean distance transform width, minimum-area suppression, calibration hook	Geometry module and measurement accuracy test (AT-06)
H5	WP-5 Decision layer and material profiles	Accept / reject / review rules, fracture derivation from crack geometry, threshold profiles per material, image-quality gate	Decision engine and threshold configuration
H6	WP-6 Edge packaging	ONNX export, int8 quantisation, float32 reference comparison, latency and size measurement, offline runtime hardening	Quantised model and performance report (AT-17, AT-18, AT-22)
H7	WP-7 Interface, batch reporting and audit log	Live inspection view, region review panel, history and filters, CSV and PDF export, append-only audit log	Working interface and batch report
H8	WP-8 Acceptance testing and documentation	Execute AT-01 to AT-22, triage and fix defects, per-material metric report, final documentation and demonstration	Acceptance test results and Phase 2 submission

14.3 Distribution of work among team members

Member	Primary responsibility	Secondary responsibility	Owned deliverables
Kanishk Rungta	Model training and export; post-processing and geometry engine; edge packaging, quantisation and performance measurement; overall technical integration	Project coordination, risk tracking, gate reporting and code review	Baseline checkpoint and metric report (WP3); geometry module (WP4); quantised model and performance report (WP6)

Member	Primary responsibility	Secondary responsibility	Owned deliverables
Nethi Kushala Kumar	Dataset consolidation and governance; annotation workflow and QA gates; decision layer and material threshold profiles; interface, batch reporting and audit log	Acceptance test execution and documentation	Dataset manifest and split-freeze report (WP1); data pipeline (WP2); decision engine and thresholds (WP5); interface and batch report (WP7)
Both, jointly	Acceptance testing, defect triage, final report and demonstration	Cross-review of each other's work packages at every gate	Acceptance test results and Phase 2 submission (WP8)

14.5 Risks and mitigation

Risk	Impact	Mitigation
Scarce real factory imagery for non-steel materials	High	Stage materials by data availability; validate steel first; use synthetic data for training only and report each material's status honestly
int8 quantisation degrades thin-defect recall	High	Compare against the float32 reference model at every export (AT-22); fall back to float32 deployment if the loss exceeds tolerance
Crack and scratch confused with each other	High	Curated hard negatives, per-class connected components, class-weighted loss, and separate per-class metrics
Over-flagging on textured or reflective surfaces	Medium	False-positive-area metric gated at 0.5%, minimum-area suppression, controlled illumination and a dedicated clean-surface test set
Two-person team capacity	Medium	Scope frozen after WP1; E-11 and E-12 treated as stretch items; one active workstream per person
Camera rig unavailable for pilot capture	Medium	Batch mode over stored images provides an alternative acceptance evidence path; calibration tests use printed reference targets
Label noise from assisted annotation	Medium	Mandatory human verification before training under FR-11, with the reviewer and outcome recorded in the manifest
Schedule slip at an early gate	Medium	Gates M1 to M4 each have a defined criterion; a missed gate cuts scope from the stretch items rather than shortening acceptance testing

15. Scalability and Future Enhancement Scope

Future enhancement	Prepared extension point
Additional materials and defect types	Versioned material profiles, an extensible taxonomy and a documented retraining pipeline
Conveyor or PLC integration	Trigger and decision interfaces kept separate from ML inference

Future enhancement	Prepared extension point
Multi-view or 3-D inspection	The acquisition layer can accept multiple synchronised views or depth data
Calibrated physical measurement at every station	The geometry module already converts pixels to units once calibration is supplied
Automated validation of flagged regions	Scores and geometry needed for it are already produced; it becomes available once flagged-region accuracy is characterised
Active learning	Review and override records identify valuable samples for human-approved retraining
Predictive quality analytics	Traceable batch records can later be correlated with process variables

16. Feasibility Conclusion

Dimension	Assessment
Technical	Feasible as an offline automated inspection and operator-monitoring prototype for visible cracks and scratches, with fracture derived from crack geometry. The selected model configuration already meets the size and latency targets with margin. Material-specific validation and real-device testing remain the substantive open work.
Economic	Compact CPU deployment on commodity edge hardware lowers the entry barrier relative to commercial machine-vision systems. No return-on-investment claim is made, since it would require factory-specific labour, scrap, rework and equipment costs that are not available to the team.
Operational	Automated accept, reject and review decisions remove the repetitive search load. Human personnel monitor system health, review uncertain cases, approve thresholds and handle overrides, so accountability stays with the quality function.
Schedule	Requirements, dataset governance and baseline architecture decisions are complete. Eight weeks of work packages remain, covering data consolidation, multi-class training, the decision layer, edge packaging, the interface and acceptance testing.
Ethical and governance	The system must communicate uncertainty, preserve audit records, retain human oversight, and must not be presented as a replacement for certified non-destructive testing where hidden or subsurface defects matter. Several public datasets carry non-commercial licences, which is recorded as a constraint on any later commercial use.

17. References

ID	Source	Relevance	Link
R1	M. Yung et al., "Examining the fatigue-quality relationship in manufacturing," Applied Ergonomics, 2020.	Fatigue accounted for up to 42% of the variance in quality deficits in the reported analysis.	https://doi.org/10.1016/j.apergo.2018.12.011

ID	Source	Relevance	Link
R2	A. Tjolleng et al., "Development of a human-friendly visual inspection method for painted vehicle bodies," Applied Ergonomics, 2023.	Controlled comparison of defect detectability and a fatigue proxy; small participant sample.	https://doi.org/10.1016/j.apergo.2022.103911
R3	J. Antony et al., "An exploration of organizational readiness factors for Quality 4.0," IJQRM, 2023.	Online survey of 147 senior managers across three continents.	https://doi.org/10.1108/IJQRM-10-2021-0357
R4	R. R. Ghatak and J. A. Garza-Reyes, "Investigating the barriers to Quality 4.0 adoption in the Indian manufacturing sector," IJQRM, 2024.	Models fourteen barriers and highlights standards and analytics-tool gaps.	https://doi.org/10.1108/IJQRM-09-2023-0277
R5	N. Hütten et al., "Deep Learning for Automated Visual Inspection in Manufacturing and Maintenance," Applied System Innovation, 2024.	Survey of 196 open-access publications.	https://doi.org/10.3390/asi7010011
R6	E. Cumbajin et al., "A Systematic Review on Deep Learning with CNNs Applied to Surface Defect Detection," Journal of Imaging, 2023.	Systematic review of 59 primary studies by surface type.	https://doi.org/10.3390/jimaging9100193
R7	Y. Chen et al., "Surface Defect Detection Methods for Industrial Products: A Review," Applied Sciences, 2021.	Reviews traditional and deep-learning approaches.	https://doi.org/10.3390/app11167657
R8	S. B. Jha and R. F. Babiceanu, "Deep CNN-based visual defect detection: Survey of current literature,"	Surveys modern visual-defect architectures and industrial applications.	https://doi.org/10.1016/j.compind.2023.103911

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	Computers in Industry, vol. 148, 2023.		
R9	D. Tabernik et al., "Segmentation-Based Deep-Learning Approach for Surface-Defect Detection," Journal of Intelligent Manufacturing, 2019.	Introduces the Kolektor segmentation approach and real microscopic defects.	https://doi.org/10.1007/s10845-019-01476-x
R10	J. Božič et al., "Mixed supervision for surface-defect detection," Computers in Industry, 2021.	Introduces KolektorSDD2 and evaluates supervision and annotation trade-offs.	https://doi.org/10.1016/j.compind.2021.103459
R11	P. Bergmann et al., "MVTec AD — A Comprehensive Real-World Dataset for Unsupervised Anomaly Detection," CVPR, 2019.	5,354 images across 15 categories with pixel-precise annotations.	https://www.mvtec.com/research-teaching/datasets/mvtec-ad
R12	Severstal: Steel Defect Detection, Kaggle competition dataset, 2019.	12,568 training images of steel sheet with run-length-encoded masks for four defect classes.	https://www.kaggle.com/c/severstal-steel-defect-detection
R13	K. Song and Y. Yan, NEU surface defect database (NEU-DET), Northeastern University.	1,800 grayscale hot-rolled steel strip images across six defect classes.	https://ieee-dataport.org/documents/neu-det
R14	X. Lv et al., "Deep Metallic Surface Defect Detection: The New Benchmark and Detection Network," Sensors, 2020.	Introduces GC10-DET: 3,570 grayscale images across ten defect types.	https://doi.org/10.3390/s20061562
R15	Y. Huang, C. Qiu and K. Yuan, "Surface defect saliency of magnetic tile," The	1,344 images across six categories with pixel-level masks.	https://doi.org/10.1007/s00371-018-1588-5

ID	Source	Relevance	Link
	Visual Computer, vol. 36, 2020.		
R16	ISO 9001:2015, Quality management systems — Requirements.	Current quality-management standard status and revision timeline.	https://www.iso.org/standard/62085.html
R17	ISO/IEC 23894:2023, Artificial intelligence — Guidance on risk management.	Guidance for AI-related risk management.	https://www.iso.org/standard/77304.html
R18	NIST AI Risk Management Framework.	Voluntary trustworthy-AI risk framework.	https://www.nist.gov/itl/ai-risk-management-framework
R19	Cognex VisionPro Deep Learning.	Industrial AI defect-inspection product information.	https://www.cognex.com/en/products/machine-vision-software/visionpro-deep-learning
R20	KEYENCE vision inspection systems overview.	Cameras, lighting, lenses, software and production-line inspection.	https://www.keyence.com/products/vision/resources/vision-resources/what-are-vision-inspection-systems.jsp
R21	MVTec Deep Learning Tool.	Labelling, training and evaluation workflow for industrial image processing.	https://www.mvtec.com/products/deep-learning-tool
R22	Landing AI manufacturing page.	LandingLens manufacturing visual-inspection positioning.	https://landing.ai/industries/manufacturing